Vedavarshita Nunna

(215) 252-6799 varshitaveda13@gmail.com veda-varshita VedaVarshita Portfolio

Experience

BioNtech Jan 2026 – present
Gen AI Researcher - Contract | Python, Databricks *Berkeley Heights, NJ*

- Designed a query-adaptive retrieval fusion strategy for an agentic RAG + knowledge graph system over a 3,285-document oncology corpus, dynamically weighting vector, graph, and keyword signals based on inferred query intent to handle heterogeneous unstructured biomedical data
- Developed a two-pass LLM entity extraction pipeline with windowed cross-chunk deduplication and content-addressed entity identity (18.5K entities, 82K+ edges on Amazon Neptune), enabling idempotent, replayable knowledge graph construction at corpus scale
- Evaluated infrastructure tradeoffs (Neptune over ArangoDB, custom async orchestration over LangGraph) under production constraints; reduced full-corpus LLM reprocessing cost 98% ($15K→$300) via prompt-version-aware extraction caching with automated cross-store integrity verification (Postgres, Databricks Delta, Neptune)

CODED Lab, Drexel University July 2024 – Dec 2025
AI/ML Research Assistant – Advisor: Dr. Jake Williams) | Pytorch, LLM *Philadelphia, PA*

- Developed a novel Bayesian Initialization algorithm using co-occurrence statistics and product quantization, improving accuracy from ∼10% baseline to 74% on MNIST and 65% on CIFAR-10; validated via 75% cosine similarity between BI-initialized and fully trained kernels, confirming efficient warm-start convergence
- Fine-tuned a 1.5B parameter LLM using RLHF with preference-based DPO on Anthropic's HHH dataset; boosted task performance 17% and cut perplexity 20% (31.64→25.42), delivering aligned, coherent outputs adhering to Responsible AI practices

Precisely July 2022 – June 2023
Associate Software Engineer - Location Intelligence R&D | Python, SQL *Bangalore, India*

- Improved and productionized global location data pipelines, expanding country coverage and streamlining processing for 1M+ records, reducing runtime by 67% and ensuring more consistent, reliable data across regions

Precisely Jan 2022 – June 2022
Intern - Location Intelligence R&D | TensorFlow, SQL, Python *Bangalore, India*

- Built an address deduplication system using Bi-LSTM with GloVe embeddings (TensorFlow/Keras), achieving 86% accuracy across diverse address formats; de-duplicated 1M+ records (40% reduction) and optimized SQL test pipelines, improving data accuracy 6% and cutting validation time 14%

Education

Drexel University Philadelphia, PA
Master of Science, Artificial Intelligence & Machine Learning (GPA: 3.9) *Sep 2023 – June 2025*

Lovely Professional University Punjab, India
Bachelor of Technology, Electronics & Communication Engineering – Minor in Data Science *July 2018 – May 2022*

Projects & Research

Synthetic Text Data Generation Pipeline with LoRA [Link] | *Pytorch, LoRA* June 2025 – July 2025

- Built a 4-stage LLM synthetic-data pipeline (zero/one/few-shot + LoRA fine-tuning) with Prefect orchestration, Docker containerization, and FastAPI serving; reduced perplexity 10–15× with validated quality metrics (ROUGE-L, Self-BLEU, VendiScore)
- Enabled reproducible experiments across 10+ hyperparameters (temperature, top-k/p, shots, LoRA rank), improving output uniqueness and coherence while cutting compute by training only adapter weights

LLaMA Vector Search Engine [Link] | *LangChain, Retrieval-Augmented Generation* Aug 2024 – Sept 2024

- Architected a semantic research assistant using LangChain and LLaMA 3.2 over 30+ ML papers; implemented FAISS vector retrieval with OllamaEmbeddings achieving 86% MRR and 56% search latency reduction
- Optimized RAG with DSPy prompt learning and MMR retrieval, improving answer relevance 27% and reducing generation latency to 1.06s; extended with a stateful LangGraph workflow achieving 94% query success via conditional routing and automated document grading

Technical Skills

LLMs & Gen AI: LLMs, SLMs, Multi-modal Models, RAG, Finetuning, Prompt Engineering, Statistical Modeling
Programming & Dev Tools: Python, C/C++, Java, NumPy, Pandas, NLP, SQL, Git, Docker, Spark, AWS
ML/AI Tools: Scikit-Learn, PyTorch, TensorFlow, OpenCV, HuggingFace, LangChain, MLflow, FAISS, Pinecone
Certifications: Introduction to Machine Learning using TensorFlow – Udacity | Data Science for Engineers – NPTEL
Achievements: Location Intelligence Hackathon Runner up (Precisely), CCI Dean's Fellowship (Drexel)